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| **Fiscal Year:** | FY 2025 | **Task Last Updated:** | FY 01/08/2025 |
| **PI Name:** | Strock, Nicole Ph.D. | | |
| **Project Title:** | Temporal Changes in Astronauts' Muscle and Cardiorespiratory Physiology Pre, During, and Post Spaceflight | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | (1) **Aerobic**:Risk of Reduced Physical Performance Capabilities Due to Reduced Aerobic Capacity<br>(2) **Muscle**:Risk of Impaired Performance Due to Reduced Muscle Size, Strength and Endurance | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | nicole.c.strock@nasa.gov | **Fax:** | FY |
| **PI Organization Type:** | GOVERNMENT | **Phone:** | |
| **Organization Name:** | KBR | | |
| **PI Address 1:** | JSC B21 / 1216F | | |
| **PI Address 2:** | | | |
| **PI Web Page:** | | | |
| **City:** | Houston | **State:** | TX |
| **Zip Code:** | 77058 | **Congressional District:** | 36 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C |
| **Start Date:** | 06/07/2021 | **End Date:** | 09/30/2033 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Brocato, Becky | **Contact Phone:** | |
| **Contact Email:** | becky.brocato@nasa.gov | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | Per the Principal Investigator (PI), Michael Bishop, Ph.D. is no longer a Co-Investigator on the project (Ed., 5/3/24). New PI as of March 2023 is Dr. Nicole Strock. Previous PIs were Dr. Eric Rivas -- and prior to Dr. Rivas, the PI was Dr. Meghan Downs. Collaborators are Jason Norcross, M.S., and Taylor Schlotman, Ph.D. for their roles in CIPHER Egress Fitness. | | |
| **COI Name (Institution):** | Prejean, Brian ( KBR/NASA Johnson Space Center )<br>Norcross, Jason ( KBR/NASA Johnson Space Center )<br>Schlotman, Taylor ( KBR/NASA Johnson Space Center )<br>Marshall-Goebel, Karina ( NASA Johnson Space Center ) | | |
| **Grant/Contract No.:** | Internal Project | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | NOTE: Continuation of "Temporal Changes in Astronauts' Muscle and Cardiorespiratory Physiology Pre, During, and Post Spaceflight" with new Principal Investigator, Dr. Nicole Strock.<br>Quantification of astronauts' changes in cardiorespiratory fitness and muscle performance in parallel with monitoring of pre and in-flight lifestyle habits (i.e., exercise training, nutritional intake, and sleep patterns) is needed to develop countermeasures and technologies for monitoring and mitigating crew health and performance risks during exploration class missions. The research proposed herein will profile temporal changes in astronauts' cardiorespiratory fitness and muscle mass, strength, and endurance over the course of spaceflight missions ranging from 2 months, 6 months, and up to 1 year in duration. Additionally, a statistical based extrapolation will provide predictions for changes associated with exploration missions 2-3 years in duration. To accomplish these objectives astronauts will be asked to participate in pre, in, and postflight measurement of muscular and cardiorespiratory fitness. Additionally, data collected by other "Complement of Integrated Protocols for Human Exploration Research" (CIPHER) investigators and Medical Operations testing (i.e., function testing, bone, dietary, exercise, sleep, and cognition outcomes) will be obtained via data sharing. Muscular fitness will be tested pre, in, and postflight using isometric (mid-thigh pull, leg press) and isokinetic (knee, ankle, trunk extension/flexion) methods. Cardiorespiratory fitness and related parameters will be tested pre, in, and postflight using aerobic capacity (VO2peak) testing. This proposal specifically addresses the temporal effects of spaceflight on changes in cardiorespiratory fitness and muscle function, both critical parameters in maintaining the ability to perform mission critical tasks and enabling safe human space exploration beyond low-Earth orbit. Integration of data collected during the pre, in, and postflight periods will lead to a better understanding of how to optimize exercise and non-exercise countermeasures to maintain crew health, safety, and performance during the exploration era. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | It is well established that exercise has many health benefits. This research will improve our understanding about the variability fitness improvements and how they change over time. Understanding this will improve exercise prescriptions for athletic and clinical populations on Earth. |
| **Task Progress:** | In this reporting period, we have coordinated with the Research Operations and Integration element to continue data collection and consenting of crewmembers for participation in this investigation. Seven total participants have signed informed consents, 4 participants are in active data collection, and 3 participants have completed data collection. |
| **Bibliography Type:** | Description: (Last Updated: 04/09/2026) |
| **Abstracts for Journals and Proceedings** | Strock N, Prejean B, Varanoske A, Norcross J, Schlotman T, Marshall-Goebel K. "Temporal changes in astronauts' muscle and cardiorespiratory physiology before, during, and after spaceflight." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025.<br>Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 |